

Explainable Hybrid Machine Learning for Multi-Class Disease Classification: A Study of Anemia CBC Data and Large-Scale Thyroid Diagnosis

Aditya Dhull

Independent Study in Applied Machine Learning

adityadhull@iisc.ac.in

Abstract—Automated classification of hematologic and endocrine disorders from routine laboratory panels can assist clinical decision support, but reported systems often rely on small single-source datasets, neglect class imbalance and missing values, and provide only global explanations. This paper presents a unified, reproducible, and explainable framework and evaluates it in two experiments. *Experiment 1* applies a hybrid Random Forest (RF) and Extreme Gradient Boosting (XGBoost) soft-voting framework to nine-class anemia classification from Complete Blood Count (CBC) data ($n=1,232$ after de-duplication). *Experiment 2* extends the same pipeline to a substantially larger and harder problem: seven-class thyroid-disease diagnosis from the Garavan Institute archive ($n=9,172$), which contains severe class imbalance (73.8% majority) and informative missing laboratory values. The framework adds stratified five-fold cross-validation, macro-averaged metrics, one-vs-rest ROC-AUC, paired significance testing, and—crucially—*detailed per-prediction* SHapley Additive exPlanations (SHAP). On anemia the hybrid attains a hold-out accuracy of 0.984 and macro-F1 of 0.947. On thyroid it attains accuracy 0.945 and macro-F1 0.863. We find that the hybrid’s improvement over the RF baseline is *not* statistically significant on anemia ($p=0.82$), and we further show, via a seed-stability analysis, that a five-fold paired t -test p -value is itself unstable (range 0.03–0.99); on the $7\times$ larger thyroid data the analogous improvement *is* significant ($p=0.014$), illustrating that statistical power—not a better model—drives the difference. SHAP explanations align with established physiology and, in one case, expose a data-quality error, demonstrating the practical value of instance-level interpretability.

Index Terms—Anemia, thyroid disease, XGBoost, Random Forest, soft voting, SHAP, explainable AI, class imbalance, macro-F1, statistical significance.

I. INTRODUCTION

Routine laboratory panels such as the Complete Blood Count (CBC) and the thyroid function panel are inexpensive, ubiquitous, and information-rich. Machine learning (ML) over such structured data can support the differential diagnosis of disorders whose subtypes carry distinct clinical management. However, three recurring weaknesses limit the trustworthiness of reported systems: (i) reliance on small, single-institution datasets; (ii) inadequate treatment of class imbalance and missing values; and (iii) explanations restricted to global feature importance, without per-patient justification.

This work is organized around a single, leakage-safe pipeline applied to two diseases. We first *apply* a hybrid RF + XGBoost soft-voting framework to nine-class anemia classification, with a leakage-safe preprocessing, evaluation protocol, and SHAP-based interpretation. We then *extend* the identical pipeline to a larger, more complex thyroid-disease problem deliberately chosen to stress the three weaknesses above. Our contributions are:

- A careful, open anemia classification study reporting an *honest* negative result on statistical significance.
- An extension to seven-class thyroid diagnosis on 9,172 patients with explicit handling of severe imbalance and informative missingness.
- Detailed *per-prediction* SHAP narratives that decompose individual predictions into clinically interpretable contributions, including for mis-classified cases.
- A seed-stability analysis showing that a five-fold paired t -test p -value is intrinsically noisy, and a demonstration that the hybrid’s advantage becomes statistically detectable only with a larger sample.

II. RELATED WORK

Classical anemia classification follows morphological reasoning on the mean corpuscular volume (MCV) and the hemoglobin (HGB) concentration, distinguishing microcytic, normocytic, and macrocytic anemias, refined by MCH and MCHC. Recent ML studies report high accuracy with tree ensembles on CBC data, and growing use of SHAP for interpretability. The hybrid framework studied here couples RF and XGBoost via soft voting, applies SHAP to the boosted component, and—unlike much prior work—tests whether the hybrid’s gain over a baseline is statistically significant.

Thyroid-disease classification from the Garavan archive is a long-standing benchmark. Its laboratory features (TSH, T3, TT4, T4U, FTI) map directly to the hypothalamic–pituitary–thyroid feedback loop, making it an ideal substrate for verifying that learned attributions match physiology. Its size, imbalance, and structured missingness make it markedly more challenging than the anemia panel, which is precisely why we adopt it for the extension.

TABLE I
DATASET CHARACTERISTICS.

	Anemia (CBC)	Thyroid (Garavan)
Patients	1,232	9,172
Classes	9	7
Features	14 numeric	6 lab + 14 binary + 2 nominal
Majority class	26.2%	73.8%
Missing values	none	yes (informative)

III. DATASETS

Table I summarizes both datasets. The anemia CBC dataset comprises 14 numeric features and nine diagnostic classes; we remove exact duplicate records (a standard de-duplication step) leaving 1,232 patients. The thyroid dataset (thyroid0387) comprises 9,172 records; we map the raw diagnosis codes into seven clinically coherent groups (negative, hypothyroid, hyperthyroid, binding-protein anomaly, non-thyroidal illness, replacement therapy, discordant results). Thyroid features mix six continuous laboratory values, 14 binary clinical-history flags, and two nominal categoricals, and contain substantial missingness (e.g. T3 missing in 28% of records). Crucially, this missingness is *informative*—a test is ordered when clinically suspected—so we retain it via missing-indicator features rather than discarding it.

IV. METHODOLOGY

A. Preprocessing

Let the dataset be $D = \{(x_i, y_i)\}_{i=1}^N$ with $x_i \in \mathbb{R}^d$ and $y_i \in \{1, \dots, K\}$. Numeric features are median-imputed with an appended binary missing-indicator; nominal features are most-frequent imputed and one-hot encoded. Because tree ensembles are invariant to monotone feature scaling, no standardization is applied. All transformers are fit *inside* the cross-validation loop so that no validation-fold statistics leak into training.

B. Hybrid Ensemble and Extensions

Random Forest reduces variance by averaging T decorrelated trees,

$$f_{\text{RF}}(x) = \frac{1}{T} \sum_{t=1}^T h_t(x), \quad (1)$$

while XGBoost minimizes a regularized objective that controls bias and complexity,

$$\mathcal{L} = \sum_{i=1}^N \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k). \quad (2)$$

Each base learner m emits posterior probabilities $P_m(c | x)$, and the hybrid predicts via *soft voting* over $M=2$ learners,

$$\hat{y} = \arg \max_{c \in \{1, \dots, K\}} \frac{1}{M} \sum_{m=1}^M P_m(c | x). \quad (3)$$

Both base learners share a single preprocessing transform, so they receive identical features. As extensions we additionally

evaluate LightGBM and a *stacking* ensemble (RF, XGBoost, LightGBM with a logistic meta-learner), which *learns* a combination rule rather than using the fixed equal weights of (3). Imbalance is optionally addressed via class weighting (RF, LightGBM) and balanced per-sample weights (XGBoost).

C. Evaluation Protocol

We use a stratified 80/20 hold-out and stratified five-fold cross-validation on the training partition, reporting the mean and standard deviation across folds,

$$\sigma = \sqrt{\frac{1}{4} \sum_{j=1}^5 (S_j - \mu)^2}. \quad (4)$$

Metrics include accuracy,

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(y_i = \hat{y}_i), \quad (5)$$

the macro-averaged F1-score,

$$F1_i = \frac{2 \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}, \quad \text{Macro-F1} = \frac{1}{K} \sum_{i=1}^K F1_i, \quad (6)$$

and one-vs-rest ROC-AUC. Macro-averaging weights every class equally and is the headline metric, since on a 73.8%-majority dataset accuracy alone is misleading.

D. Statistical Significance

To test whether the hybrid genuinely outperforms the RF baseline, we apply a paired t -test to the fold-wise macro-F1 scores,

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}, \quad (7)$$

where d_i is the per-fold difference, $\bar{d}$ its mean, s_d its standard deviation, and $n=5$. Because both models are evaluated on identical folds, the test is paired, removing fold-to-fold variance. We complement it with a seed-stability study (Section V-C).

E. Explainability

For a tree model and a target class, SHAP decomposes a prediction additively,

$$f(x) = \phi_0 + \sum_{j=1}^d \phi_j, \quad (8)$$

where ϕ_0 is the base value and ϕ_j is feature j 's signed contribution. We use exact TreeSHAP on the XGBoost component to obtain global importances (mean $|\phi_j|$) and, for each patient, an ordered list of contributions that we render as a human-readable narrative and a waterfall plot.

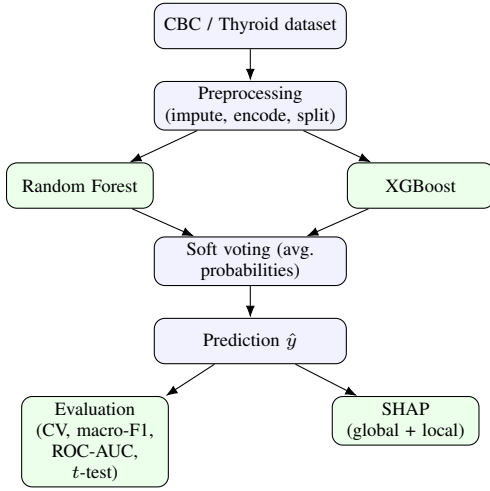


Fig. 1. The shared, leakage-safe pipeline used for both experiments.

TABLE II
EXPERIMENT 1 (ANEMIA): STRATIFIED FIVE-FOLD CROSS-VALIDATION,
MEAN $\pm$ STD.

Model	Accuracy	Macro-F1	ROC-AUC
Random Forest	0.985 $\pm$ 0.013	0.934 $\pm$ 0.055	0.9996
XGBoost	0.988 $\pm$ 0.004	0.948 $\pm$ 0.008	0.9994
Hybrid (RF+XGB)	0.987 $\pm$ 0.007	0.939 $\pm$ 0.017	0.9998
LightGBM	0.989 $\pm$ 0.006	0.947 $\pm$ 0.053	0.9997
Stacking	0.991 $\pm$ 0.006	0.966 $\pm$ 0.025	0.9998

V. RESULTS

A. Experiment 1: Anemia

Table II reports five-fold cross-validation. The hybrid attains 0.987 accuracy and 0.939 macro-F1. On the hold-out test set the hybrid attains accuracy 0.984, macro-F1 0.947, and ROC-AUC 0.9997 (Table V). Per-class F1 exceeds 0.94 for all well-populated classes; the sole weak class is Macrocytic anemia (F1 0.67 on three test samples), a small-sample artifact. Notably, the *stacking* extension achieves the best cross-validated macro-F1 (0.966), the one configuration that clearly surpasses the soft-voting hybrid.

The paired *t*-test of hybrid versus RF yields $\bar{d}=+0.0053$, $t=0.24$, $p=0.82$: *not* significant. On roughly 1,200 patients the hybrid’s edge over a strong baseline is within fold noise.

B. Experiment 2: Thyroid (Extension)

Table III reports cross-validation on the thyroid data. The hybrid attains 0.952 accuracy and 0.875 macro-F1; on the hold-out set it attains accuracy 0.945, balanced accuracy 0.850, macro-F1 0.863, and ROC-AUC 0.995 (Table V). The gap between accuracy (0.945) and macro-F1 (0.863) is the imbalance story in one line: the model is excellent on large classes and merely good on rare ones. Per-class results (Table IV) show near-perfect detection of hypothyroidism (F1 0.967) but difficulty on the intrinsically ambiguous discordant-results

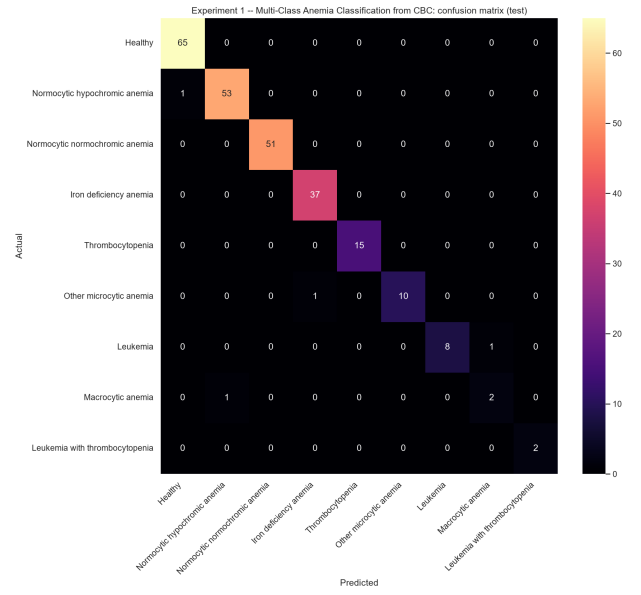


Fig. 2. Experiment 1: confusion matrix of the hybrid on the anemia hold-out set.

TABLE III
EXPERIMENT 2 (THYROID): STRATIFIED FIVE-FOLD CROSS-VALIDATION,
MEAN $\pm$ STD.

Model	Accuracy	Macro-F1	ROC-AUC
Random Forest	0.941 $\pm$ 0.008	0.850 $\pm$ 0.024	0.992
XGBoost	0.950 $\pm$ 0.006	0.872 $\pm$ 0.020	0.995
Hybrid (RF+XGB)	0.952 $\pm$ 0.005	0.875 $\pm$ 0.018	0.995
LightGBM	0.951 $\pm$ 0.005	0.873 $\pm$ 0.018	0.994
Stacking	0.951 $\pm$ 0.005	0.874 $\pm$ 0.015	0.991

class (F1 0.633)—a class defined by *disagreeing* assays, where no clean signal exists.

Here the paired *t*-test of hybrid versus RF yields $\bar{d}=+0.0255$, $t=4.16$, $p=0.014$: *significant*. The same kind of improvement that was undetectable on 1,200 anemia patients becomes detectable on 9,172 thyroid patients—a direct illustration of statistical power.

C. p-Value Stability

Because $n=5$, the paired *t*-test is high-variance. Holding the data and models fixed and varying only the fold-shuffling seed, the hybrid-versus-RF p-value on anemia ranges from 0.03 to 0.99 (Table VI); nine of ten seeds are non-significant, one crosses 0.05 by chance. Our 0.82 is an unremarkable draw from this distribution. The robust outcome is therefore the *verdict* (not significant), not any single p-value—a caution against over-interpreting one statistic from few folds.

VI. EXPLAINABILITY ANALYSIS

A. Global Attribution

For anemia, the top global SHAP features are HGB, MCV, WBC, MCHC, MCH, and PLT, recovering the MCV-centered

TABLE IV
EXPERIMENT 2 (THYROID): PER-CLASS HOLD-OUT PERFORMANCE OF THE HYBRID.

Class	Prec.	Rec.	F1	Support
negative	0.963	0.974	0.969	1355
hypothyroid	0.949	0.985	0.967	132
nonthyroidal illness	0.977	0.923	0.949	91
binding protein	0.829	0.808	0.818	78
replacement therapy	0.903	0.915	0.909	71
discordant results	0.738	0.554	0.633	56
hyperthyroid	0.804	0.789	0.796	52
macro avg	0.880	0.850	0.863	1835

TABLE V
HOLD-OUT TEST PERFORMANCE OF THE HYBRID IN BOTH EXPERIMENTS.

Metric	Anemia	Thyroid
Accuracy	0.984	0.945
Balanced accuracy	0.939	0.850
Macro-F1	0.947	0.863
Weighted-F1	0.984	0.944
ROC-AUC (OvR)	0.9997	0.995

morphological framework that underpins hematologic diagnosis. For thyroid, the top features are TSH, T3, FTI, TT4, on-thyroxine, and T4U—exactly the thyroid function panel, ordered as a clinician would expect (Fig. 3). The missing-indicator features for T3 and TSH also rank highly, confirming that *which* tests were ordered carries diagnostic signal.

B. Per-Prediction Explanation

Equation (8) lets us narrate individual predictions. For a thyroid patient predicted *hyperthyroid* (Fig. 4), the dominant contributions are a very high free thyroxine index (FTI = 196) and a suppressed TSH (0.02)—the textbook signature of an overactive gland under negative feedback. For an anemia patient predicted *iron deficiency*, low MCV and low MCH drive the decision, matching the microcytic–hypochromic phenotype. Because contributions sum to the model output, each explanation is a complete account rather than a post-hoc rationalization.

C. Explaining Errors and Data Quality

SHAP applied to mis-classified cases is diagnostically useful. One anemia patient labeled *Normocytic hypochromic* but predicted *Healthy* carries physiologically impossible values (HGB = 41, RBC = 13.1, HCT = 2); the SHAP decomposition shows the absurd HGB drove the error, turning a silent misclassification into a visible, diagnosable data-quality problem. This exemplifies interpretability as a debugging tool, not merely a trust signal.

VII. DISCUSSION

Three findings stand out. First, the anemia results are strong and stable, including an honest negative significance result.

TABLE VI
ANEMIA HYBRID-VS-RF PAIRED *t*-TEST UNDER DIFFERENT FOLD SEEDS.

Seed	mean(Hyb–RF)	p-value	Verdict
13	+0.0002	0.990	not sig.
42	+0.0053	0.824	not sig.
0	+0.0067	0.762	not sig.
21	+0.0184	0.425	not sig.
7	+0.0204	0.118	not sig.
99	+0.0325	0.029	significant
Range over 10 seeds: [0.029, 0.990]; significant in 10%.			

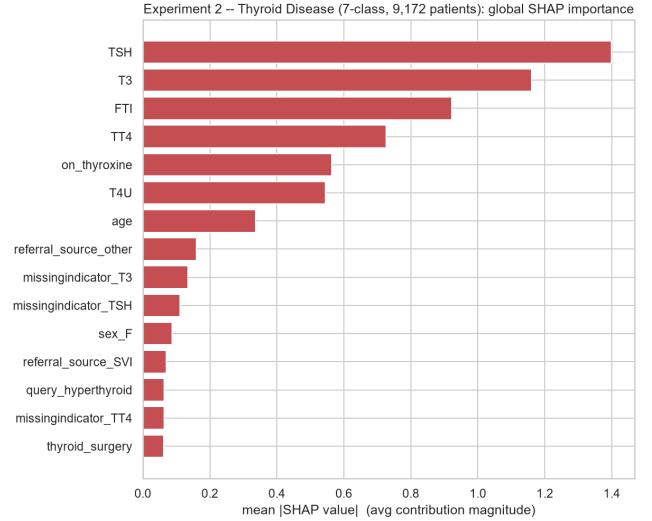


Fig. 3. Experiment 2: global SHAP importance (mean $|\phi_j|$) of the XGBoost component on the thyroid data.

Second, scaling to a $7\times$ larger, imbalanced, missingness-laden thyroid problem is feasible with the same pipeline, but the macro-F1 drops to 0.863—not because the model is worse, but because the problem is harder and the evaluation is more honest (an intrinsically ambiguous class and severe imbalance). Third, the significance contrast ($p=0.82$ on anemia versus $p=0.014$ on thyroid), together with the seed-stability analysis, makes a methodological point that transcends either dataset: with few folds and a near-zero effect, a p-value is noise, and only sample size makes a small true effect detectable. Throughout, SHAP explanations agree with physiology and occasionally reveal data problems, supporting their inclusion in clinical decision-support pipelines.

VIII. LIMITATIONS

Both datasets are single-institution and, for thyroid, decades old; external multi-center validation is needed. The thyroid class grouping is a modeling choice that merges fine-grained diagnoses. Hyperparameters were only lightly tuned, and SHAP provides associational, not causal, attributions. Minority classes (Macrocytic anemia; discordant thyroid results) remain under-represented, limiting the reliability of their per-class estimates.

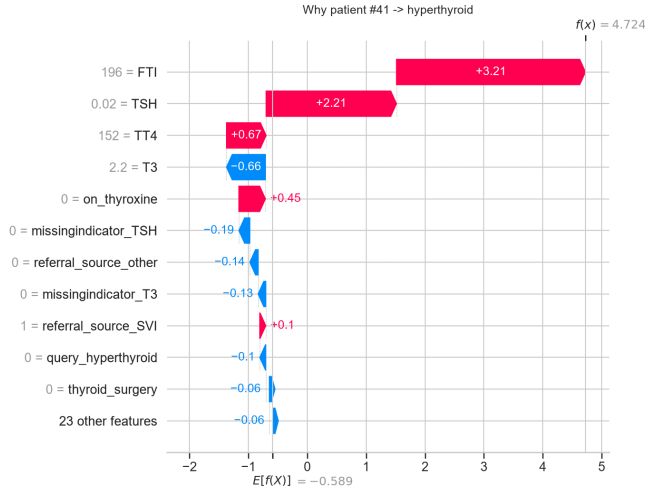


Fig. 4. Experiment 2: per-prediction SHAP waterfall for a patient correctly predicted as hyperthyroid; FTI and suppressed TSH dominate.

IX. CONCLUSION

We presented a reproducible, explainable hybrid ML framework and validated it on two multi-class disease problems. Experiment 1 is a nine-class anemia classifier with an honest non-significant model comparison. Experiment 2 extends the pipeline to a larger, harder seven-class thyroid problem, explicitly handling imbalance and informative missingness, and shows that the hybrid’s advantage becomes statistically significant only with more data. Detailed per-prediction SHAP explanations align with established physiology and can surface data-quality errors. Future work includes external validation, cost-sensitive learning for rare classes, SHAP interaction analysis, and probability calibration for clinically actionable confidence estimates.

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